Nonparametric Density Estimation II

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Outline

- Nonparametric Density Estimation
- Histogram Approach
- Parzen-window method
- K_n-Nearest-Neighbor Estimation
- Gaussian Mixture Models
- Expectation and Maximization

Interim Summary

Histogram

From histogram to density estimation

Convergence conditions

Parzen window

Parzen window Function (Kernel)

Cube kernel, Gaussian kernel

Window size and performance

Classification

K_n – Nearest Neighbor

$$p(\mathbf{x}) \approx \frac{k/n}{V} \qquad (*)$$

$$\lim_{n \to \infty} V_n = 0, \qquad \lim_{n \to \infty} k_n = \infty, \qquad \lim_{n\to\infty} \frac{k_n}{n} = 0$$

$$\lim_{n \to \infty} p_n(x) = p(x).$$

$k_n = \sqrt{n}$

$V_n \approx 1/(\sqrt{n}\,p(x))$

$$p_n(x) = \frac{k_n/n}{V_n}$$

Adaptive window size

K_n – Nearest Neighbor

$V_n = 1/\sqrt{n}$

$$k_n = \sum_{i=1}^{n} \varphi\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n}\right)$$

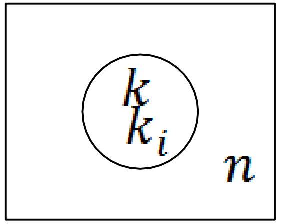
$$p_n(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{V_n} \varphi\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n}\right).$$

Same window size

Parzen window

Classification with K-NN and Parzen window: Estimation of a posteriori probabilities

- The k-NN (and Parzen window) techniques can be used to estimate the a posteriori probabilities $p(\omega_i|\mathbf{x})$ from a set of n labeled samples.
- Suppose that we place a cell of volume V around x and capture k samples:
  - k_i are labeled ω_i
  - $(k - k_i)$ have other labels.



- A simple estimate for the joint probability density $p(\mathbf{x}, \omega_i)$ is

$$p_n(\mathbf{x}, \omega_i) = \frac{k\ /n}{V}$$

- A reasonable estimate for $p(\omega_i|\mathbf{x})$ is

$$p_n(\omega\ |\mathbf{x}) = \frac{p_n(\mathbf{x},\omega_i)}{\sum_{j=1}^{C} p_n(\mathbf{x},\omega_j)} = \frac{k_i}{k}$$

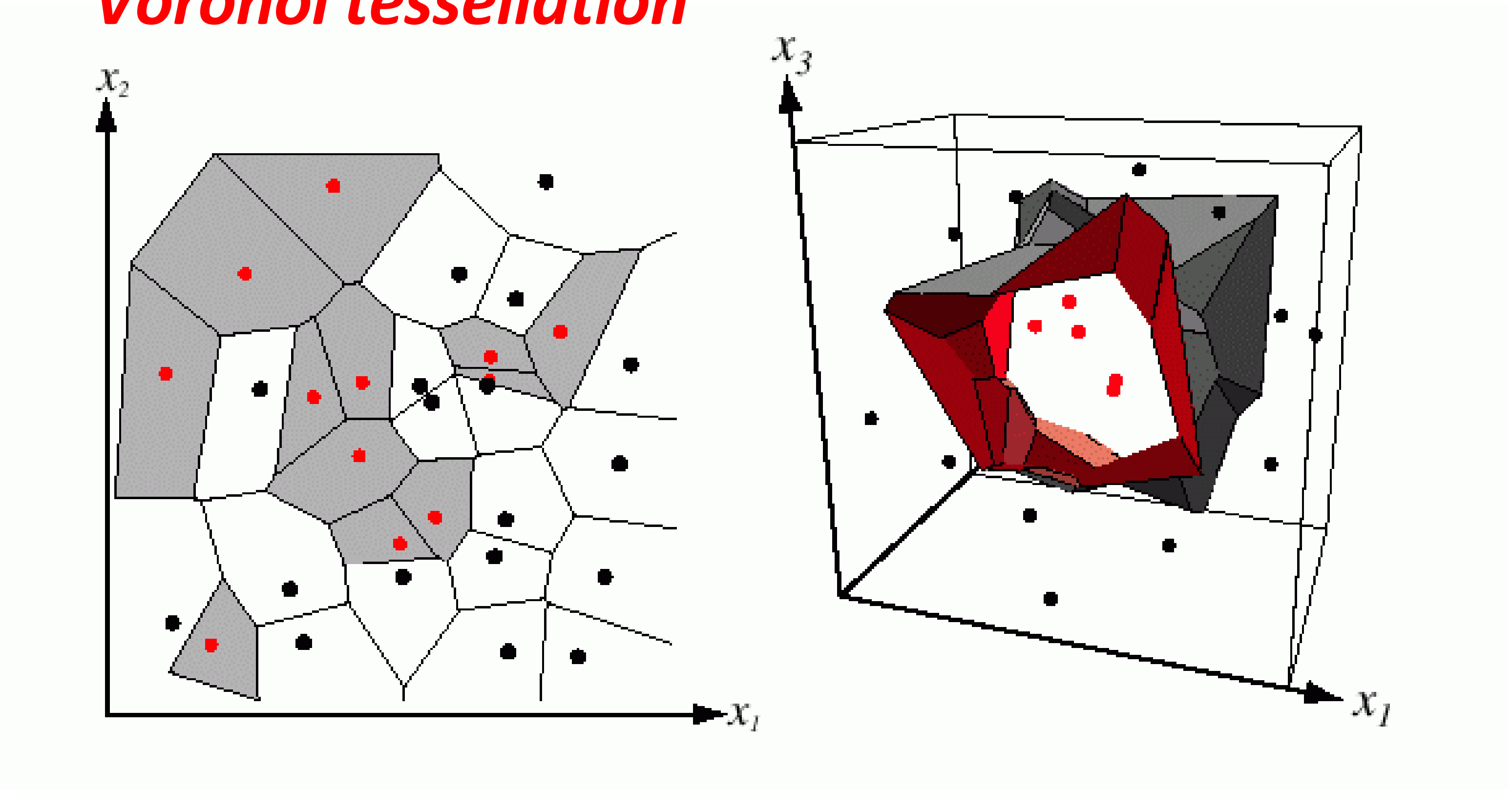
Instance-Based Learning Classifier
Approximated Minimum error classifier

Classification With Nearest Neighbor Rule

- Let $D^n = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ denote a set of n labeled prototypes and let $\mathbf{x}' \in D^n$ be the prototype nearest to a test point $\mathbf{x}$.
- Then the Nearest Neighbor Rule: assign the label of $\mathbf{x}'$ to $\mathbf{x}$.
- This rule is suboptimal, but when the number of prototypes is large, its error is never worse than twice the Bayes rate.

Classification With Nearest Neighbor Rule

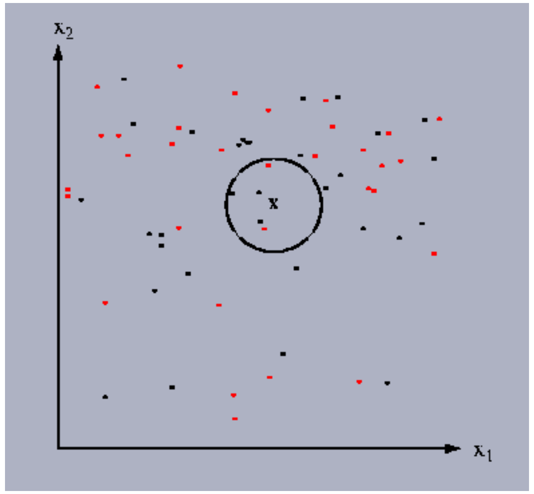
Voronoi tessellation



Voronoi cells

Classification With the k-Nearest Neighbor Rule

- The k-NN query starts at the test point and grows a spherical region until it encloses k training samples, and it labels by a majority vote of these samples.
- **Algorithm**:
  - For each sample point Compute Distance (sample point, test point)
  - Sort the distances
  - Inspect the k smallest distances
  - Label test point by a majority vote.



$$p_n(\omega_i|\mathbf{x}) = \frac{p_n(\mathbf{x},\omega_i)}{\sum_{j=1}^{C} p_n(\mathbf{x},\omega_j)} = \frac{k_i}{k}$$

Classification With the k-Nearest Neighbor Rule

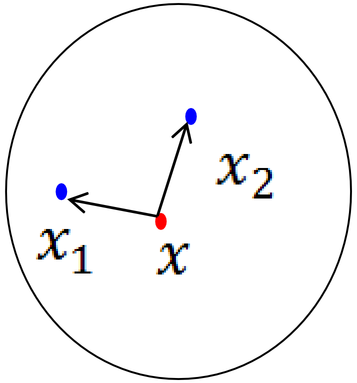
- Question.
  - If a *posteriori* probabilities $p(\omega_i|\boldsymbol{x}),\ i = 1, 2$ for two classes are known, for example

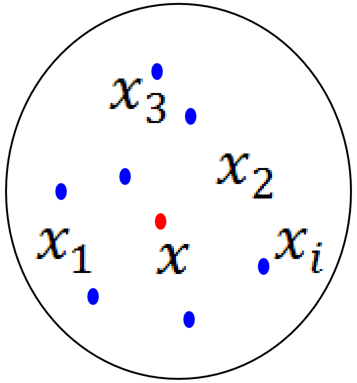
$$p(\omega_1|\mathbf{x}) > p(\omega_2|\mathbf{x})$$

  - What is a probability of choosing a class ω_1 for x with the Bayes, the nearest neighbor (NN), the k-NN classifiers?

Classification With the k-Nearest Neighbor Rule.

- **Answer**
  - Bayes: always w_1
  - NN: $p(\omega_1|x)$
  - K-NN: $\sum_{i=(k+1)/2}^{k} \binom{k}{i} p(\omega_1|\mathbf{x})^i (1-p(\omega_1|\mathbf{x}))^{k-i}$





Exercise

On the coast, the probability of catching a salmon is 0.6, and the probability of catching a tuna is 0.4.
The probability that a salmon is less than or equal to 40cm in size is 0.2, and the probability that a tuna is less than or equal to 40cm in size is 0.03.
Let we catch a fish and its size is 40cm or less, calculate the probability of classifying this fish as a salmon using Bayes, NN, and K-NN (K=9) classifiers.

Solution

P(salmon) = 0.6, P(tuna) = 0.4

P(less than 40cm| salmon) = 0.2, P(less than 40cm| tuna) = 0.03

Posteriori probability P(salmon |less than 40cm) and P(tuna |less than 40cm) are given as:

$$p(tuna|less\ than\ 40cm) = \frac{p(less\ than\ 40cm|tuna)p(tuna)}{p(less\ than\ 40cm)} = \frac{0.2 \times 0.6}{0.6 \times 0.2 + 0.4 \times 0.03} = 0.909$$

$$p(salmon|less\ than\ 40cm) = \frac{p(less\ than\ 40cm|salmon)p(salmon)}{p(less\ than\ 40cm)} = \frac{0.03 \times 0.6}{0.6 \times 0.2 + 0.4 \times 0.03} = 0.0909$$

Using posterioris above, we have:

(1) In Bayes classifier case, we always classify this fish as a salmon (100%).
(2) In the case of nearest neighbor (NN), we classify this fish with a P(salmon|less than 40cm)=0.909 probability.
(3) In the case of K=9, we calculate

$$\sum_{i=5}^{9} \binom{9}{5} p(salmon|less\ than\ 40cm)^{i} (1 - p(salmon|less\ than\ 40cm)^{k-i}$$

Therefore we have:

$$14 \times 0.909^5 \times 0.0909^4 + 84 \times 0.909^6 \times 0.0909^3 + 36 \times 0.909^7 \times 0.0909^2$$
$$+9 \times 0.909^8 \times 0.0909^1 + 0.909^9 \times 0.0909^0 = 0.9937837$$

We classify this fish with probability 0.994.

Gaussian Mixture Models (GMM)

Gaussian mixture models can be used for –

- Anomaly detection: identifying fraud or detecting errors in data collection
- Probabilistic Labeling
- Generate synthetic data points for data augmentation

In Real-world Scenario

- Finding patterns in medical datasets
- Modeling natural phenomena
- Customer behavior analysis
- Stock price prediction: by volatility and noise detection
- Gene expression data analysis

Goal

Density Estimation (in our case, GMM) with:

- Expectation-Maximization algorithm
- Sampling Method
- Variational Inference

First, let's try to estimate the parameter of the likelihood (even this is challenging).

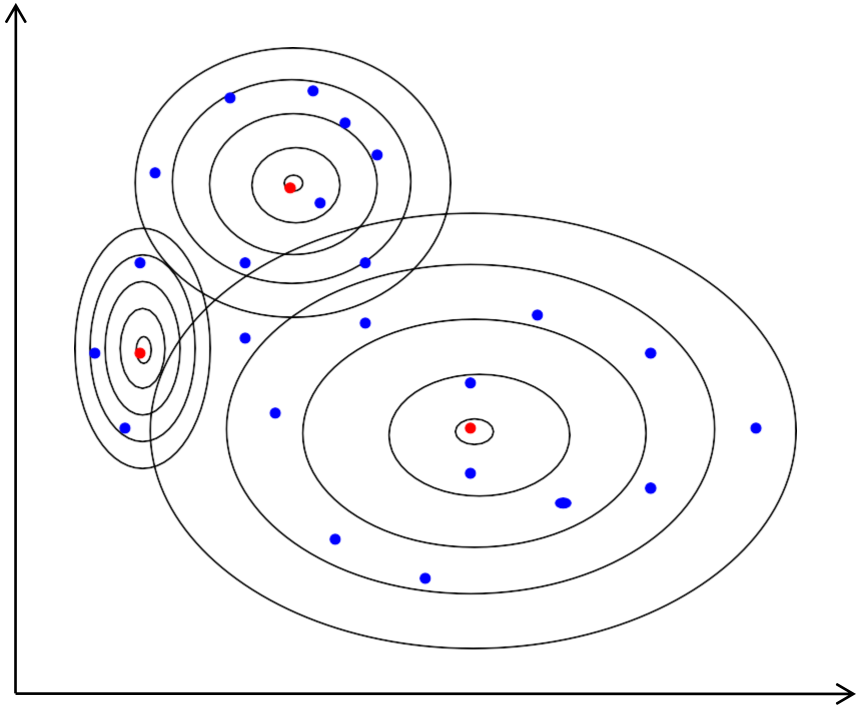
Gaussian Mixture Estimation

- Parzen Window

$$p_n(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{V_n}\phi\left(\frac{\mathbf{x}-\mathbf{x}_i}{h_n}\right).$$

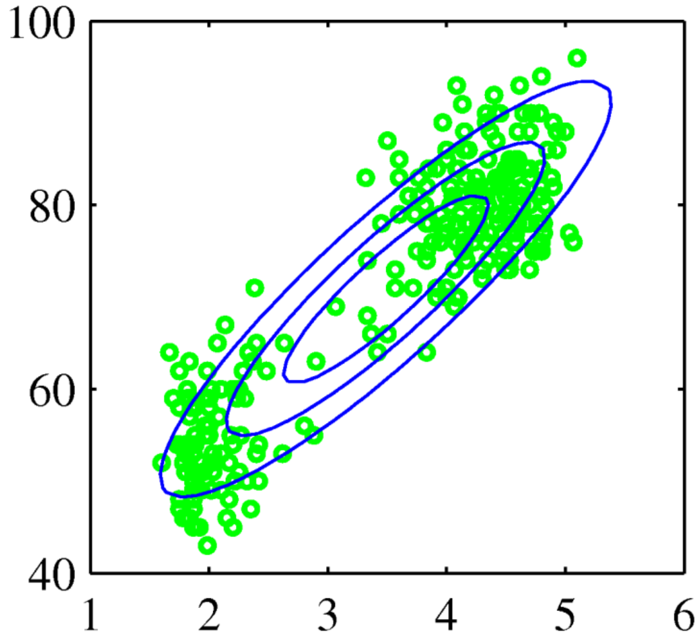
- Gaussian Mixture

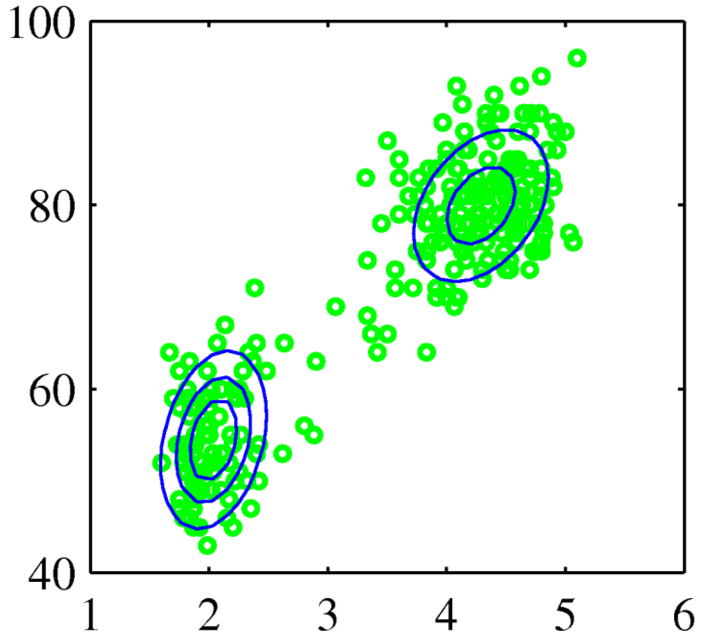
$$p(\mathbf{x}) = \sum_{k=1}^{K} w_k \ \varphi\left(\frac{\mathbf{x}-\mu_k}{\sigma_k}\right).$$



Gaussian Mixture Estimation

- While the Gaussian distribution is commonly used, the real world can be far from straightforward.
- In practice, there exist cases where applying Gaussian assumptions may not be realistic.
- Consider the examples of such cases as the following:

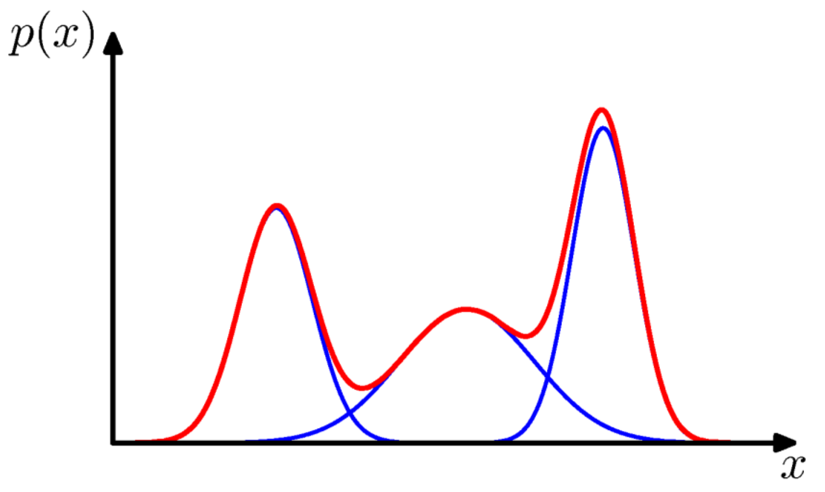




Gaussian Mixture Estimation

• In order to address this challenge, we now introduce Gaussian Mixture Model (GMM).

• Gaussian mixture model allows for the representation of very complex densities
• By using a sufficient number of Gaussian distributions
• By adjusting not only the coefficients of the linear combination but also the mean and covariance parameters, we can approximate nearly all continuous densities



Mixture of Gaussians

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k N(\mathbf{x}|\mu_k, \Sigma_k)$$

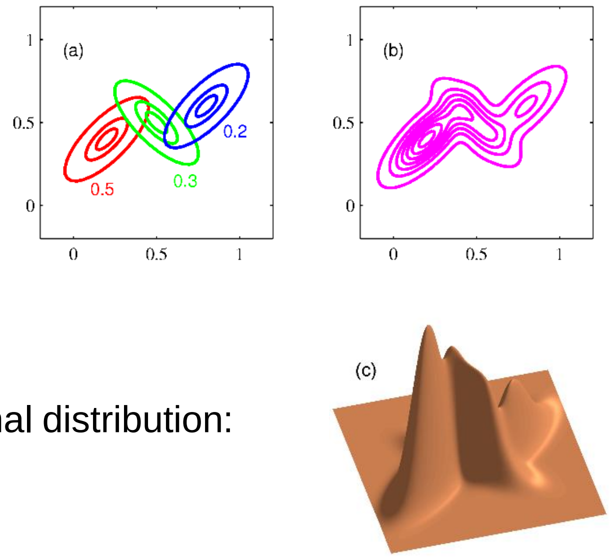
- Each Gaussian distrib. $N(\mathbf{x}|\mu_k, \Sigma_k)$: component
- Each distribution is the likelihood of the parameter

* In generalized models, it is also possible to mix two (or more than two) different distributions as well. (ex: Bernoulli mixture model)

Gaussian Mixture Estimation

- Here, π_k is the mixing coefficients.
- Each components are normalized. $p(x)$ should be normalized, so we have the following:

$$\sum_{k=1}^{K} \pi_k = 1$$



- For coefficient we have $N(x|\mu_k, \Sigma_k) \geq 0$, thus we have $p(x) \geq 0$.
- Considering the above conditions, we have:

$$0 \leq \pi_k \leq 1$$

- Suppose that we choose k^{th} component with the probability $p(k)$. Then we have the following marginal distribution:

$$p(x) = \sum_{k=1}^{K} p(k)p(x|k)$$

- This equation should be equal to the previous expression of $p(x)$.
- Thus we have $\pi_k = p(k)$, and we can consider it as the prior of the k^{th} component.
- Here, $p(x|k) = N(x|\mu_k, \Sigma_k)$.

Gaussian Mixture Estimation

- Now, consider the posteriori $p(k|x)$, which is *responsibilities (will be used later)*
- By using Bayes rule, we have this posteriori as:

$$\gamma_k(x) \equiv p(k|x) = \frac{p(k)p(x|k)}{\sum_l p(l)p(x|l)} = \frac{\pi_k N(x|\mu_k, \Sigma_k)}{\sum_l \pi_l N(x|\mu_l, \Sigma_l)}$$

- The parameters of GMM is π, μ and Σ.
- To set the optimal parameters, we can perform MLE.
- The log-likelihood of our GMM is given as:

$$\ln p(X|\pi, \mu, \Sigma) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k N(x_n|\mu_k, \Sigma_k) \right\}$$

- Here $X = x_1, x_2, \ldots, x_N$ is the training data that we sampled.
- The challenge is that the above log-likelihood for MLE does not have the closed-form solution.
- We cannot apply the derivation to have the optimal solution.

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\pi, \boxed{\mu}, \Sigma) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k) \right\}$$

Since we do not have the closed form solution, but let us just try!
Using MLE, let us try to have optimal solutions – by taking the partial derivatives of our log-likelihood according to each parameters

1) Partial derivative of μ

$$0 = -\sum_{n=1}^{N} \frac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)} \Sigma_K^{-1}(\mathbf{x}_n - \mu_k)$$

Note that $\gamma(z_{nk}) = \dfrac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)}$

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\pi, \boxed{\mu}, \Sigma) = \sum_{n=1}^{N} \ln\left\{\sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)\right\}$$

Multiply Σ_k both side we have:

$$\mu_k = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\mathbf{x}_n$$

$$N_k = \sum_{n=1}^{N}\gamma(z_{nk})$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)}}$$

- The parameter μ is estimated in a way similar to the sample mean for x.
- However, the responsibilities, i.e., the posterior p(k|x) is used as the weight.
- Thereby μ is estimated as the weighted sum.
- Thus, by the responsibilities (posterior), the extent to which one datapoint x influences the k-th Gaussian distribution is reflected to μ .

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\pi, \mu, \boxed{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k) \right\}$$

2) Similarly, taking the partial deviation of Σ_k and set to zero we have:

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk})(\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)}}$$

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\boxed{\pi},\mu,\Sigma) = \sum_{n=1}^{N} \ln\left\{\sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k,\Sigma_k)\right\}$$

3) Same for the parameter π_k

Note that we have the constraint $\sum_{k=1}^{K} \pi_k = 1$. To handle the constraint, we apply Lagrangian and we have:

$$\ln p(\mathbf{X}|\pi,\mu,\Sigma) + \lambda\left(\sum_{k=1}^{K} \pi_k - 1\right)$$

$$0 = \sum_{n=1}^{N} \frac{N(\mathbf{x}_n|\mu_k,\Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j,\Sigma_j)} + \lambda$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k,\Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j,\Sigma_j)}}$$

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\boxed{\pi}, \mu, \Sigma) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k) \right\}$$

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$$0 = \sum_{n=1}^{N} \frac{\boxed{} N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)} + \lambda$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)}}$$

multiply π_k to have $\gamma(z_{nk})$, finally we have (details are omitted):

$$\pi_k = \frac{N_k}{N}$$

$$\boxed{N_k = \sum_{n=1}^{N} \gamma(z_{nk})}$$

Gaussian Mixture Estimation

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\pi, \mu, \Sigma) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

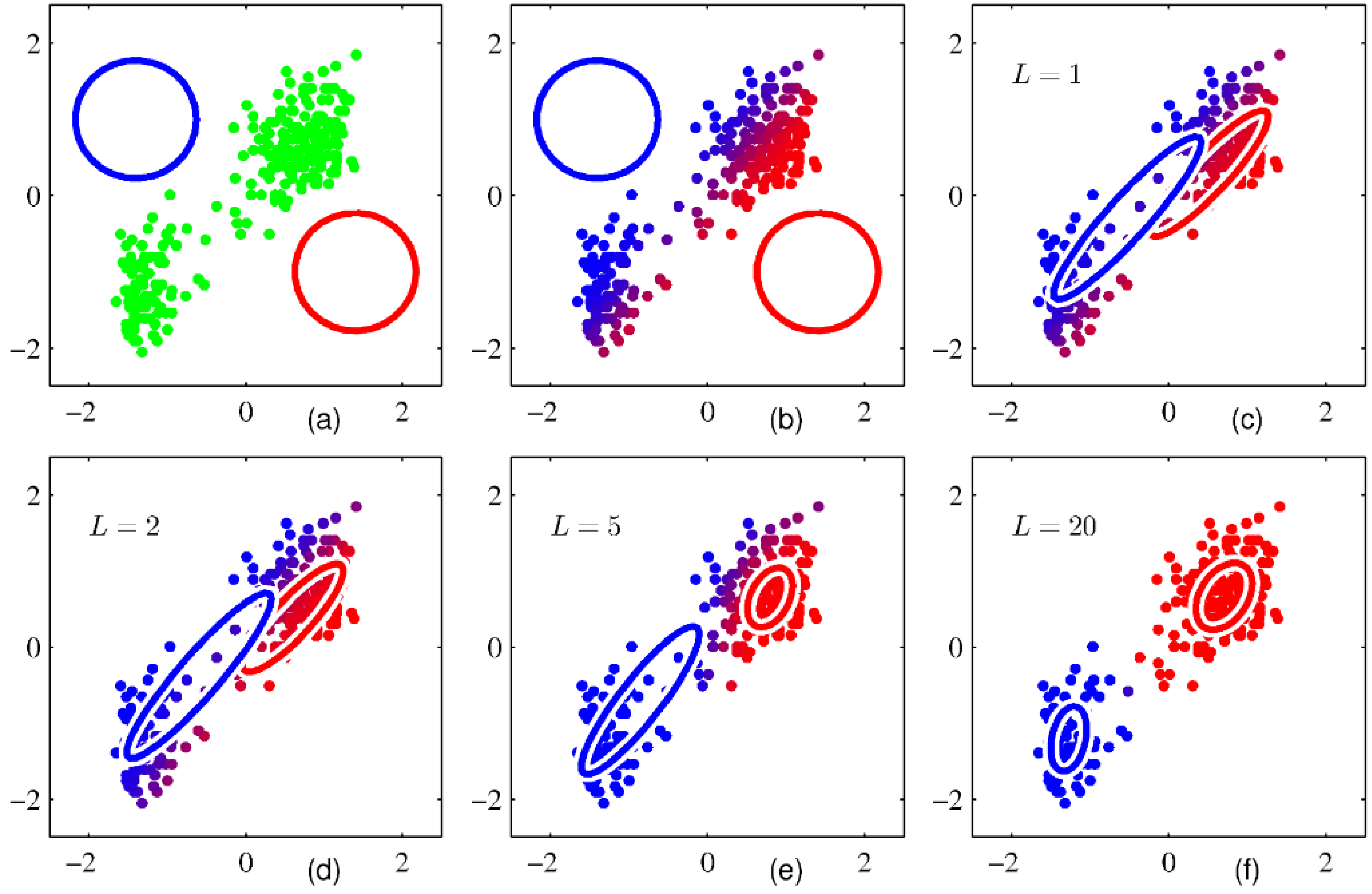
$$\boxed{N_k = \sum_{n=1}^{N} \gamma(z_{nk})}$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n | \mu_j, \Sigma_j)}}$$

In the above, solutions of each parameter are determined by other solutions simultaneously

- Do not have the closed-form solution
- Now what to do?

Expectation-Maximization for Gaussian Mixtures



1.Initialization step:
Initialize with random means and variances, as shown in figure (a) (parameter initialization).

2.Expectation step:
Use the current parameter values to calculate responsibilities (posterior probabilities).

3.Maximization step:
Utilize the calculated posterior probabilities from the Expectation step to re-estimate the parameters through Maximum Likelihood Estimation (MLE), as shown in figure (c).

4.Repeat the process, as shown in figures (d), (e), and (f).

The process of applying EM to GMM estimation

Expectation-Maximization for Gaussian Mixtures

Log-likelihood of GMM

$$\ln p(\mathbf{X}|\pi,\mu,\Sigma) = \sum_{n=1}^{N} \ln\left\{\sum_{k=1}^{K} \pi_k N(\mathbf{x}_n|\mu_k,\Sigma_k)\right\}$$

$$\mu_k = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\mathbf{x}_n$$

$$\Sigma_k = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})(\mathbf{x}_n-\mu_k)(\mathbf{x}_n-\mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

$$\boxed{N_k = \sum_{n=1}^{N}\gamma(z_{nk})}$$

$$\boxed{\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k,\Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j,\Sigma_j)}}$$

Expectation step:

Using the given parameter from the previous step, compute responsibilities

$$\gamma(z_{nk}) = \frac{\pi_k N(\mathbf{x}_n|\mu_k,\Sigma_k)}{\sum_j \pi_j N(\mathbf{x}_n|\mu_j,\Sigma_j)}$$

Maximization step:

Using precomputed responsibilities, estimate other parameters

$$\mu_k^{new} = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\mathbf{x}_n$$

$$\Sigma_k^{new} = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})(\mathbf{x}_n-\mu_k)(\mathbf{x}_n-\mu_k)^T$$

$$\pi_k^{new} = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^{N}\gamma(z_{nk})$$

Interim Summary

- K_n-Nearest Neighbor Method
- Nearest Neighbor Method
- k -Nearest Neighbor Method
- k -Nearest Neighbor Classifier
- Bayes, the nearest neighbor, the k-NN classifiers
- Gaussian Mixture Model
- EM Algorithm